

DEFENCE — FINAL PROJECT

Early detection of *high-potential* GitHub repositories.

*Anomaly detection & growth-based pseudo-
labelling on 18 months of GH Archive event data.*

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WHY EARLY DETECTION MATTERS

By the time a
repo *trends*,
the move is over.

*We forecast the breakout six months
before the trending page sees it.*

0.1 %

OF PUBLIC REPOSITORIES EVER REACH THE TRENDING PAGE

~73K

REPOSITORIES CREATED EVERY DAY ON GITHUB

6 mo

FORECAST HORIZON — FROM FIRST-YEAR ACTIVITY ALONE

GH ARCHIVE — 18 MONTHS • 10 % SAMPLE

Every public event on GitHub, *since 2011.*

SOURCE	GH Archive on BigQuery — <code>githubarchive.month.*</code>
WINDOW	Jan 2023 → Jun 2024 • 18 months
SAMPLING	Deterministic 10 % — <code>MOD(repo.id, 10) = 0</code>
EVENTS	Watch • Fork • PullRequest • Push • Issues • Create
STORAGE	Snappy-compressed AVRO on HDFS • Hive layer on top



FOUR STAGES — ONE BASH MAIN.SH

From BigQuery to Superset, *reproducibly*.



CLUSTER
hadoop-03 · hadoop-04 · YARN

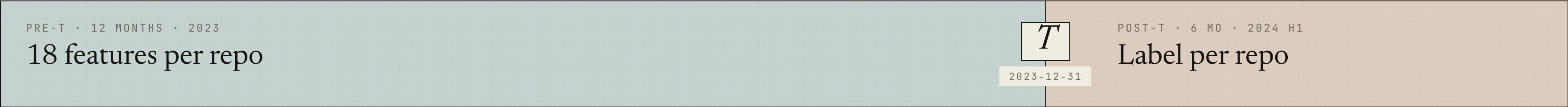
STORAGE FORMAT
AVRO · Snappy · sequential scans

REPRODUCIBILITY
end-to-end via bash main.sh

EACH STAGE OUTPUTS TO */OUTPUT* — STAGE N+1 READS FROM THERE.

GENERATING LABELS FROM THE FUTURE

No human labelled *anything*. The future labels itself.



```
success = 1 when post_watches ≥ 10 and post_watches ≥ 2 ×  
pre_watches and pre_watches > 0 = 0 otherwise
```

METHODOLOGICAL NOTE

WatchEvent $\equiv$ **star**. GitHub renamed Watch → Star around 2014; the API event name stayed WatchEvent.

Threshold calibrated empirically from q3: registration (500, 3×) collapsed the positive class to **0.04 %** — too rare for stable AUPR training. The chosen (10, 2×) yields a workable **0.08 %**.

— Borges & Valente, "What's in a GitHub Star?" (MSR 2018).

STAGE 1 — POSTGRESQL + SQOOP → HDFS

CSV → relational → *distributed.*

HOST	hadoop-04 • team28_projectdb
SCHEMA	repositories — repo_id, repo_name, first_seen_at, language events — event_type, repo_id, event_date, event_count, unique_actors
LOADER	psycopg2 COPY FROM STDIN • ~ 44 M rows
OUTPUT	AVRO + Snappy under project/warehouse

SQOOP • IMPORT-ALL-TABLES

```
# Pull both tables into HDFS as Snappy AVRO
sqoop import-all-tables \
  --connect          jdbc:postgresql://hadoop-04/team28_projectdb \
  --username         team28 --password-file .pwd \
  --as-avrodatafile \
  --compression-codec snappy \
  --warehouse-dir    project/warehouse \
  --m 1
```

Generated .avsc + .java schema files land in output/ — Stage 2 picks them up to declare the external Hive tables.

STORAGE — PARTITIONED + BUCKETED

A small Hive, *tuned for joins.*

EVENTS_PART

PARTITIONED BY (year, month)

AVRO + Snappy • clean partition pruning on every date filter

REPOSITORIES_BUCK

CLUSTERED BY (repo_id) INTO 16

AVRO + Snappy • co-located join keys for events × repos

SIX EDA QUERIES

One question. One answer. One chart.

- 01

Top language ecosystems

where breakout candidates concentrate
- 02

Monthly star dynamics

baseline rhythm of attention, 2023 vs 2024
- 03

Pre-T vs post-T growth distribution

just how rare the success class is
- 04

Success rate by activity level

does early activity actually matter?
- 05

Behavioural cohorts — success vs rest

what successful repos do differently
- 06

Q4 2023 acceleration anomalies

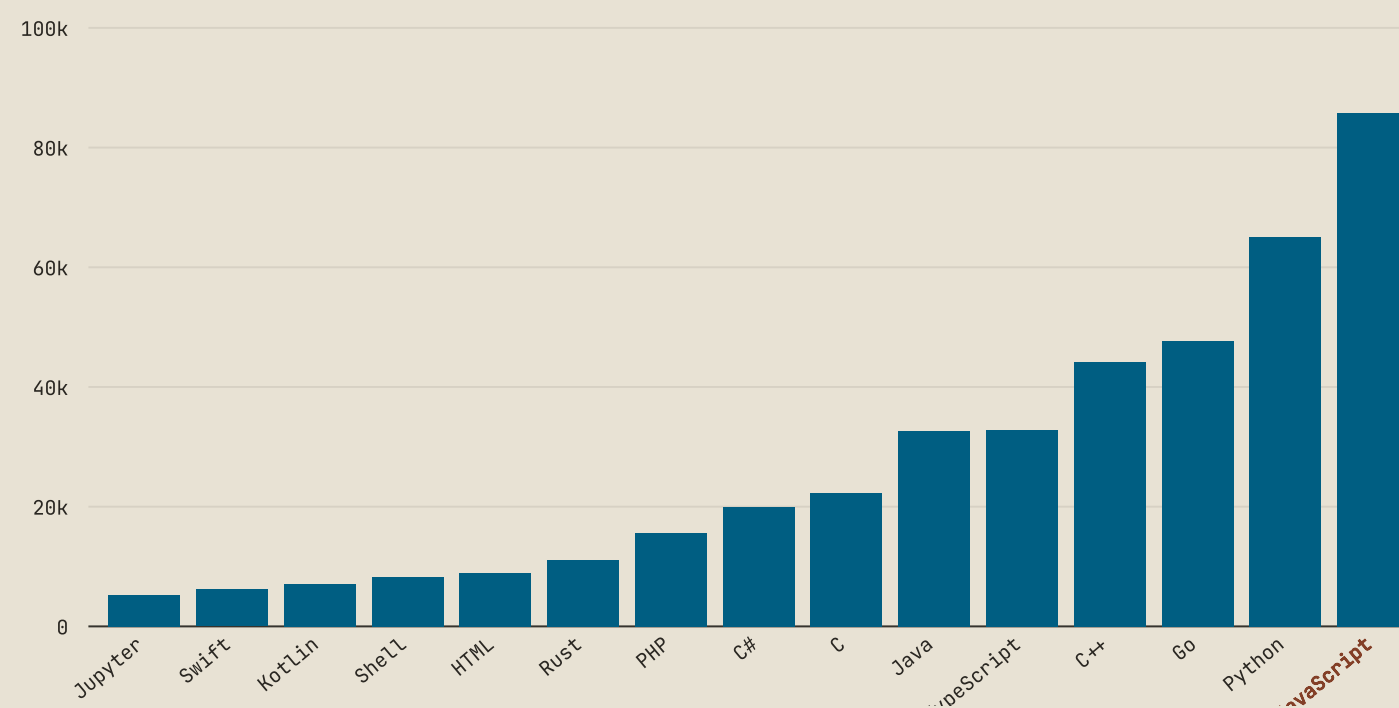
the kind of signal the model must catch

Q1 + Q2

Where stars come from, *and when*.

Q1 • TOP LANGUAGE ECOSYSTEMS

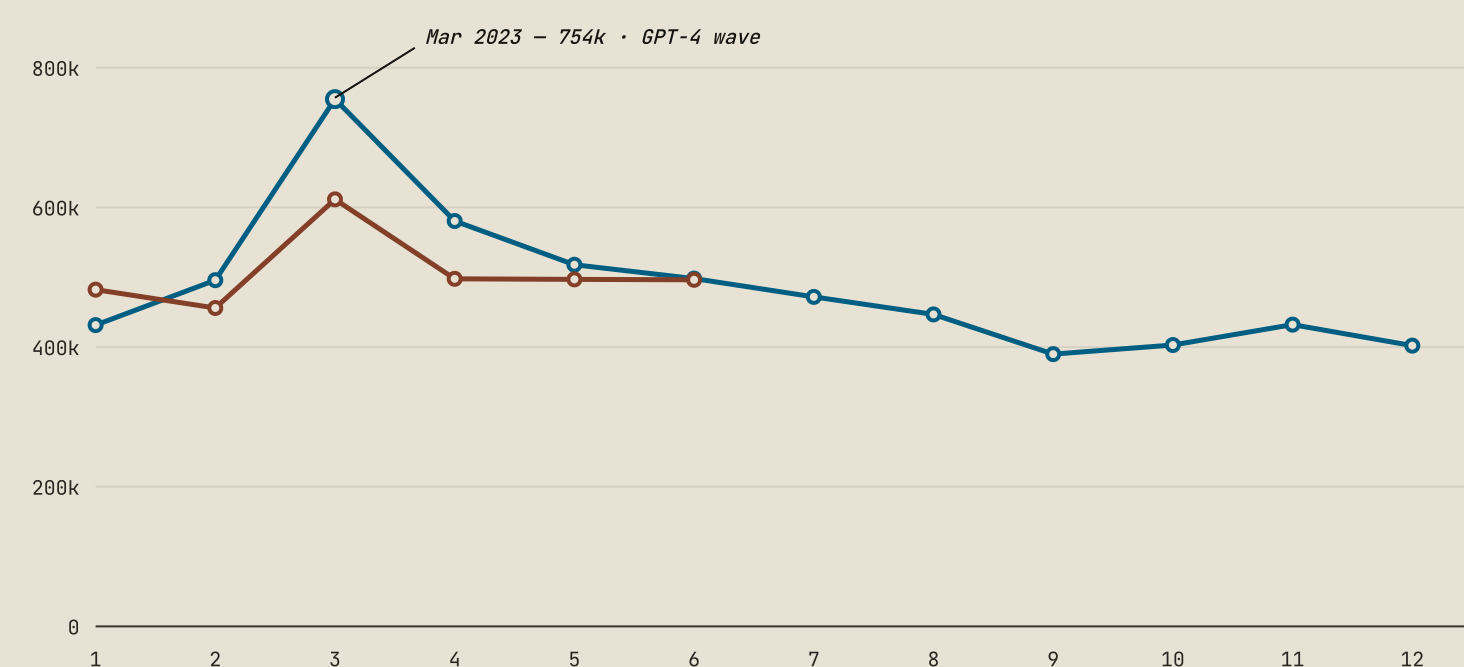
SUM(PRE-T STARS)



JS & Python lead by absolute volume — but per-repo *candidate density* peaks in **JavaScript**, **Python**, **Go**..

Q2 • MONTHLY WATCHEVENTS

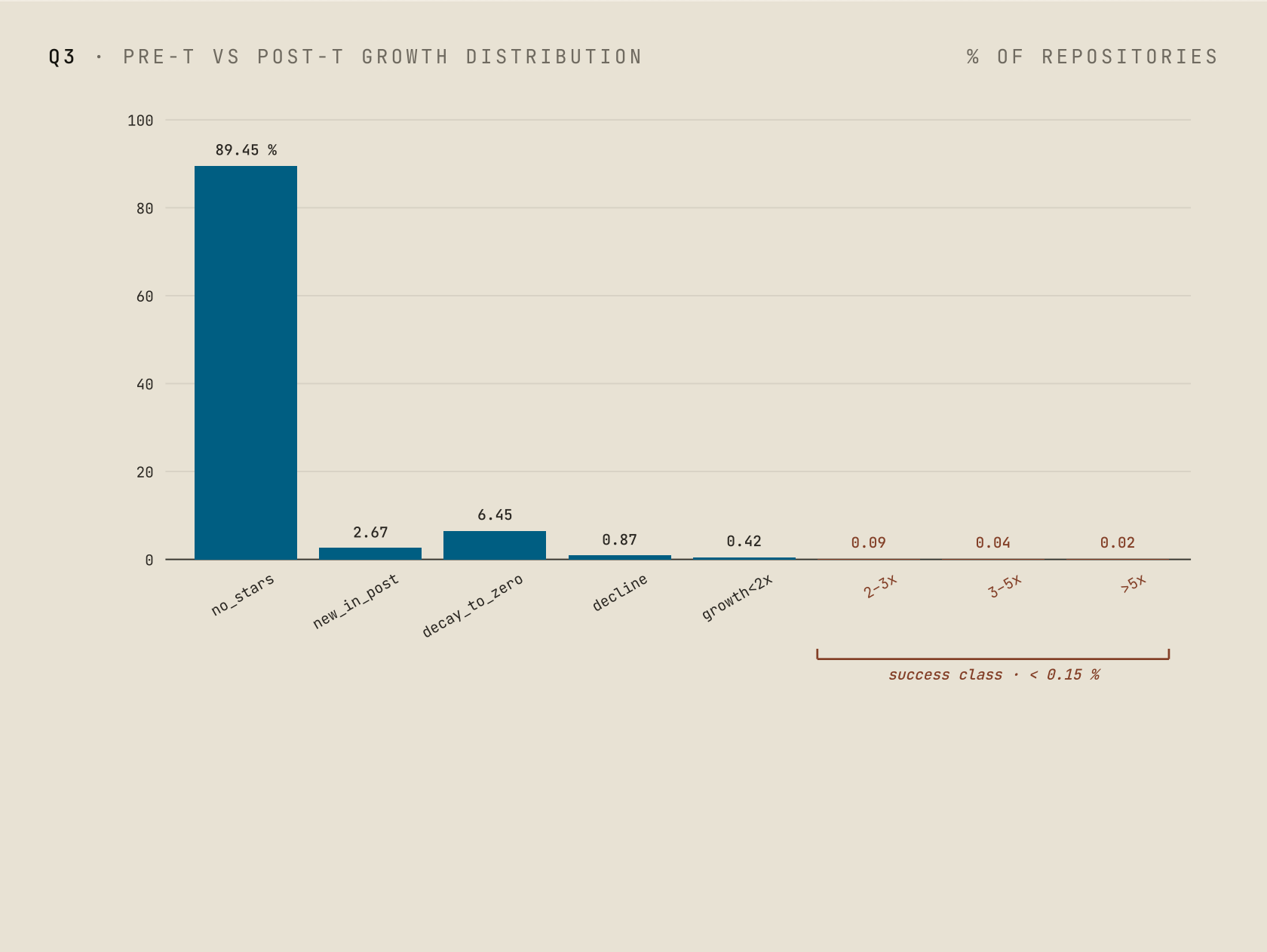
2023 2024



Baseline $\approx$ 430–500k stars per month. **March 2023 spike** $\rightarrow$ **754k** aligns with the GPT-4 / open-LLM launch wave.

Q3 + Q5

Rare. *And different.*



Q5 • BEHAVIOURAL PROFILE — SUCCESS VS REST			AVERAGE PER REPO	
METRIC	NON-SUCCESS	SUCCESS	LIFT	
events	12	285	×	23
stars	0.8	85	×	110
push share	0.37	0.33	≈	1
watch share	0.07	0.41	×	6
PR share	0.04	0.08	×	2
issues share	0.01	0.07	×	7
create share	0.49	0.07	÷	7
fork share	0.02	0.05	×	2.5

Successful repos are **established** projects, not freshly *created* ones — engagement metrics dominate over creation.

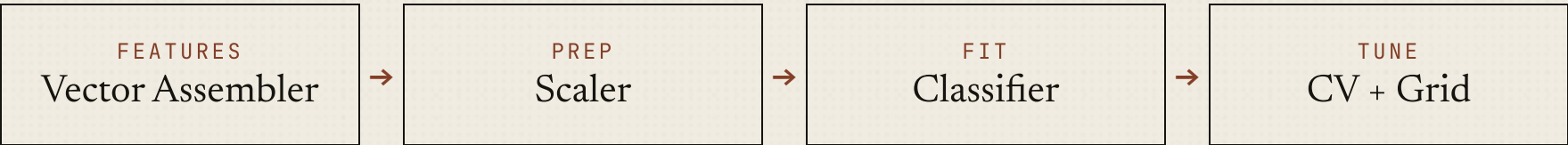
THREE CLASSIFIERS • GRID + CV • YARN

One pipeline, *three families.*

18 features • stratified 70/30 split → train 1.47 M • test 630 K. Each PipelineModel wraps assembler + scaler + classifier, so inference loads a single artefact.

SELECTOR	3-fold CrossValidator • areaUnderPR
CLUSTER	spark-submit --master yarn
PERSISTED	best model + per-test predictions on HDFS

PIPELINEMODEL — PER CLASSIFIER



CLASSIFIER	SCALER	TUNED HYPER-PARAMETER
Random Forest	MinMaxScaler	numTrees • maxDepth
Linear SVM	StandardScaler	regParam • maxIter
Naive Bayes	MinMaxScaler	smoothing • modelType

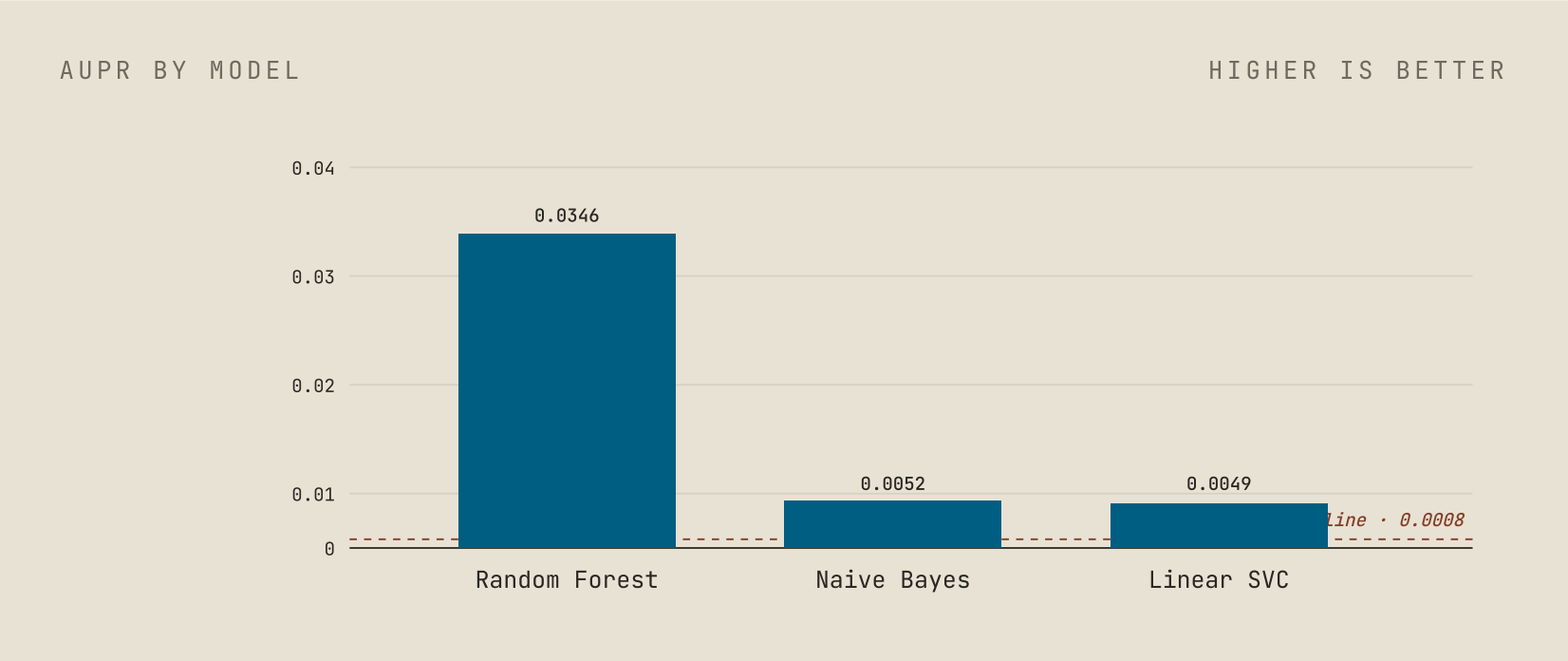
Three families — ensemble trees, geometric margin, probabilistic Bayes — for honest diversity under 0.08 % positive class.

TEST SET • 629 798 ROWS • 0.08 % POSITIVES

RF wins PR-AUC by $\approx 43\times$ over baseline.

MODEL	AUROC	AUPR	VS. BASELINE
Random Forest	0.9758	0.0346	$\times 43$
Naive Bayes	0.9166	0.0052	$\times 6.5$
Linear SVM	0.9044	0.0049	$\times 6.1$
Random PR-AUC baseline at this prevalence = 0.0008			

AUROC is high across the board because the negative class is trivially separable. Under **0.08 %** imbalance, AUPR is the only honest metric.



SAMPLE PREDICTION

repo 21331090 • true label = 1



NB flipped positive — RF / SVM played safe. The classic precision/recall trade-off at extreme imbalance.

LIVE DEMO • HADOOP-03 • PORT 8808

One narrative — *six charts.*

URL

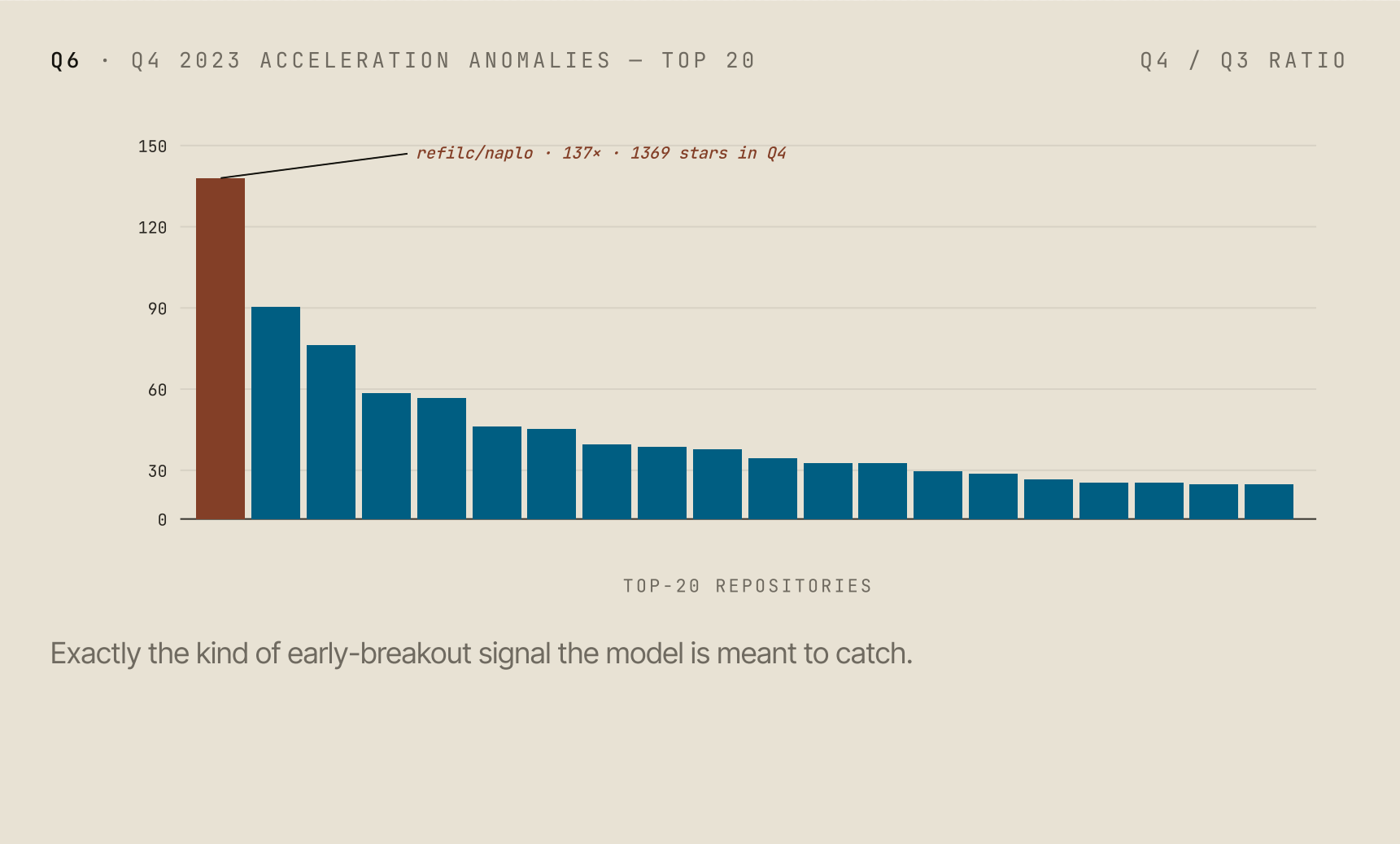
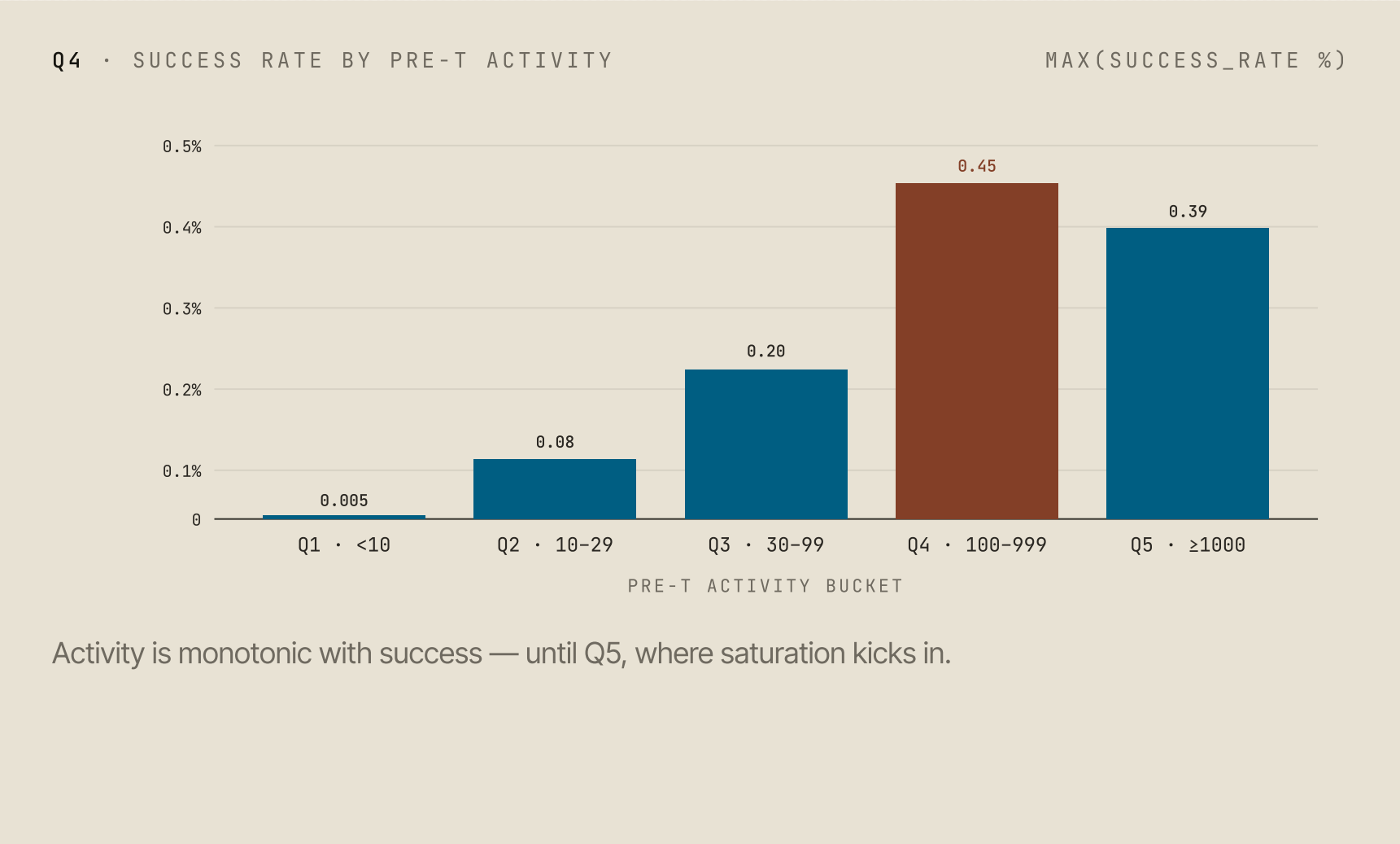
hadoop-03.uni.innopolis.ru:8808

DB

Hive • team28_projectdb

LOGIN

team28@innopolis.university



FOUR HARD PROBLEMS WE HIT ON THE CLUSTER

Where reality *broke the spec.*

01 • INGESTION

Sqoop ↔ Hive

Sqoop date types

Sqoop maps PG DATE / TIMESTAMP to AVRO long (epoch ms). Direct `CAST (... AS DATE)` failed silently.

FIX

`from_unixtime(... DIV 1000)`

02 • QUERIES

Hive 3 type-coercion

UNION ALL coerced 100 → 0 / 1

The original q4 used a `CROSS JOIN` on a thresholds matrix. Hive collapsed 100 / 250 / 500 / 1000 down to 0 / 1.

FIX

`single GROUP BY` on string activity buckets

03 • STORAGE

Tez OOM

Dynamic-partition INSERT crashed

Default 512 MB containers could not handle the 44 M-row INSERT with dynamic partitioning enabled.

FIX

`4 GB containers + sort.dynamic.partition`

04 • MODELLING

class imbalance

Positive class at 0.04 % • degenerate

Registered (500, 3×) made classifiers predict 0 for everything; AUPR couldn't separate them.

FIX

`recalibrate to (10, 2×) → 0.08 % positives`

Honest about *caveats*. Confident about the *pipeline*.

CAVEATS

- | **Stars measure attention**, not technical merit — the pseudo-label is a proxy.
- | **6-month forecast horizon** — slow-burning breakouts are out of scope.
- | **No standalone unsupervised model** — captured via `intra_growth`.
- | **Language coverage partial** — $\approx 70\%$ of top accelerators have `language = NULL`.

CONCLUSION

- | **End-to-end pipeline** — BigQuery → Postgres → HDFS → Hive → Spark ML → Superset.
- | **RF gives $\approx 7\times$ -baseline lift** in PR-AUC predicting breakout from early activity.
- | **Fully reproducible** from `bash main.sh`.
- | **Future:** isolation-forest layer · 12-month horizon · full dataset.

OPEN THE FLOOR

Questions?

METHODOLOGY
Salavat

INFRASTRUCTURE
Ildar

EDA • DASHBOARD
Anisya

FRAMING • LIMITS
Ilona